

MTH313M PRACTICE PROBLEMS WEEK 4 AND 5

Question 1. Consider a linear growth Birth and Death process $\{X_t\}_{t \geq 0}$ with $X_0 = 1$ and no immigration. Find the probability that k births happen before the first death – express in terms of the infinitesimal birth and death parameters λ and μ .

Question 2. Events, occurring according to a Poisson Process with rate $\lambda > 0$, are registered by a counter. However, each time an event is registered by the counter, it becomes inoperative for the next α unit of time and does not register any events during that time. Let $\{M_t\}_{t \geq 0}$ denote the process for the number of registered events. Find the probability that the first k events are registered. Also find $\mathbb{P}(M_t \geq k)$.

Question 3. Let $\{X_t\}_{t \geq 0}$ be an Yule process with $X_0 = N$ and with the infinitesimal birth rate $\beta > 0$. Find $\mathbb{P}(X_t \geq n)$.

Question 4. Consider an M/M/1 queueing system in the stationary state. Show that the distribution of time between successive departures has the same distribution as the inter-arrival time distribution.

Question 5. Complete the exercises left out in class.